



Evaluation of the reconnaissance results in geothermal exploration using GIS

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Abstract

Geographical Information System (GIS) is used to determine the spatial association between geophysical and geological evidence and production zones in a well-known geothermal field (Los Azufres, Mexico). Surface observations in Los Azufres were used to delineate areas characterized by high permeability and hot fluid transport from the reservoir: main faults, superficial fracture density, surface manifestations, contacts with the most recent rhyolite domes, and low values in the apparent resistivity surveys. Three knowledge-driven models were constructed based on a conceptual model of the field: a hydrothermal system in rough terrain with secondary permeability. Boolean, Index Overlay and Fuzzy scheme models were proposed and the results obtained show a good correlation with the location of the producing and non-producing wells that have been drilled in the field. The results obtained are useful for well siting (Boolean and Low-Risk Fuzzy models) or for planning further detailed exploration (Index Overlay and High-Risk Fuzzy models). © 2000 CNR. Published by Elsevier Science Ltd. All rights reserved.

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1. Introduction

Geographical Information Systems are important tools for integral interpretation of geoscientific data using a computerized approach, especially in

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exploration work. They have been used to determine the spatial association among diverse evidence sets that characterize a studied phenomenon in such a way that a prediction can be made about the possible occurrence of that phenomenon in the area of interest.

GIS models have been successfully applied in regional exploration studies of mineral resources (Bonham-Carter et al., 1988; Agterberg, 1989; Bonham-Carter, 1991; Katz, 1991; Chung et al., 1992; Bonham-Carter et al., 1994). The evidence may be analyzed also with the support of an expert system to take decisions about further work (Campbell et al., 1982; McCammon, 1993). The application of GIS in mineral exploration is based on the fact that the fundamental principles in the formation of mineral deposits are too complex to permit the application of mathematical analytical models; therefore, the knowledge of an expert is required to provide the proper weighing of the available evidence and their integration in a predictive map of mineral deposits potential (Bonham-Carter, 1991).

Geothermal exploration could use this kind of technique after the reconnaissance exploration stage, when enough data have been obtained to decide about further detailed exploration or well drilling. The accomplishment of drilling a discovery well is the end of the exploration stage and the start of exploitation. Before a well can be sited, the collection of geoscientific data usually includes surface geology, resistivity surveys, geomagnetic and gravity measurements, and geochemistry of surface manifestations. These data must be analyzed by experts who will take a decision about the best location for siting a discovery well. The complexity of active hydrothermal systems is similar to the processes that generate mineral deposits in whose exploration GISs have proved to be effective. The application of a GIS to mineral exploration has always been done in a regional setting; however, in the case that is presented here, the analysis is focused on the potential of a GIS to predict the location of the most productive areas within a single geothermal field. The application of a GIS is a further step in the evolution of computer-aided decision-taking processes in geothermal exploration. A well-known field (Los Azufres, Mexico) was used to estimate spatial relationships among all the available and relevant thematic maps and the geothermal reservoir. The Los Azufres geothermal field is already producing electricity from geothermal energy and the location of the producing areas is known well enough to perform a comparison of the results of the model and the proved occurrence of production areas.

2. Geological setting

Los Azufres geothermal field is located in the central part of the Mexican Volcanic Belt. The geologic studies indicate that it is contained in a complex caldera structure (Fig. 1). Outcrop samples show that silicic volcanic rocks are the most common in the area, and that ignimbrite layers and rhyolitic domes predominate; the regional stratigraphy of Los Azufres area has been described by Ferrari et al. (1991). Cores and cuttings from the wells drilled in the geothermal

field indicate that within the reservoir mostly andesitic rocks have been encountered with minor amounts of dacite and basalt. The andesitic rocks that form the reservoir have very low primary porosity and the permeability is mostly due to secondary porosity, so that the production areas are constrained to main active faults. Regional structures are characterized by an extension system that defines grabens and semigrabens bounded by en-echelon faults with direction E–W and NE–SW (Ferrari et al., 1991); the volcanic and sedimentary strata are tilted as a result of movement along the NNW principal fault. Local structural studies (López-Hernández, 1990) identify three fault systems: the main two systems (E–W and NE–SW) coincide with the regional trends, and the third system (N–S) is observed only in the northern part of the field (Fig. 1). The local study by López-Hernández (1991) also found that high permeability values are strictly associated with the main faults. In the southern zone (Los Azufres, Puenteillas and Agua Fria faults), the highest permeability is not restricted to the main faults but spreads away from them to distances of approximately 100 m. The E–W fault system is very important within the field; it is also the most recent (0.3–0.15 Ma) and controls the occurrence of most surface manifestations (López-Hernández, 1991). Surface manifestations are fumaroles, acid hot springs, steaming ground, mud pools, and kaolinized areas that indicate main upflows of the hydrothermal system.

The density of surface fracturing was mapped within the field (López-Hernández, 1991), and areas with a high density of fractures were observed. These fractures may be generated by shallow stresses related to hydrothermal activity and were therefore included as one of the thematic maps for the evaluation of the field.

Some areas in the field show intense fracturing but also abundant deposition of hydrothermal minerals, which causes self-sealing and a decrease in permeability. This effect is difficult to estimate without permeability tests, as hydrothermal minerals would yield low resistivity anomalies, and some thermal anomalies may still be present in those areas. These areas may constitute a source of errors when evaluating the potentially productive areas in the geothermal field.

3. Geophysical data

Geophysical studies performed in the field have included resistivity surveys, gravimetric and magnetometric maps (García, 1995). Magnetic and gravimetric data generally provide information on a regional basis. They are commonly used to define the location of intrusive bodies and regional faults. In the Los Azufres case, geomagnetic and gravity maps do not show sharp anomalies within the field, as they are dominated by regional trends. Gravity data indicate that Los Azufres geothermal field is located in a gravity minimum anomaly that may be related with a depression filled by volcanic-sedimentary strata. The local maxima in gravimetric and magnetometric maps are interpreted as being produced by intrusive bodies with higher density and magnetization than the surrounding rock

(García, 1995); these maps include mostly regional data relevant to tectonic studies (Campos-Enriquez, and Garduño-Monroy, 1995); although available, they were therefore not included in the GIS modeling of this geothermal field.

The best indicators of the production zones are the low apparent resistivity areas, which are related with the presence of some alteration minerals (mostly clays) and rock saturated with high-temperature saline fluids. CFE (Comision Federal de Electricidad) performed resistivity measurements with a Schlumberger array for several electrode separations (up to $AB/2 = 4$ km). Resistivity anomalies may be misleading due to the fact that low resistivity may be caused by the presence of hot saline water and/or clay minerals. In the case of clay minerals, they can also be present in areas of relic hydrothermal activity where, at present, no geothermal potential might be available. In order to distinguish the currently active areas from the relic hydrothermal areas at depth, information is needed on the geothermal gradient, which was not available for Los Azufres. The analysis was therefore undertaken assuming that all areas with low resistivity were prospective for geothermal exploitation.

4. Conceptual model of the field

A model of the resource to be studied is essential in order to judge the utility of the geoscientific data to provide the best information, in this case, about the geothermal production areas. From the evidence provided by surface manifestations chemistry and the geological data, the Los Azufres geothermal field can be considered as a “typical” rough topography hydrothermal system, where the manifestations on the main upflow of the system are acid-sulfate hot springs (steam-heated water) and fumaroles (Huitron et al., 1991). The rough topography favors ascent of vapor and gases derived from boiling of the geothermal fluid at depth. The mixture of vapor and gases condenses close to the surface, generating acid hot springs (due to H_2S and CO_2 oxidation). In some areas the fluids emerge in the form of fumaroles. In this type of geothermal field, there might be a spatial relationship between the acid surface manifestations and the location of the reservoir at depth.

The stratigraphic column in the area shows the predominance of volcanic rocks and the lack of porous strata with high primary permeability (Ferrari et al., 1991; Huitron et al., 1991), as would be the case for Cerro Prieto (Pelayo et al., 1991), which indicates that the reservoir permeability must be related to fracturing. This fact increases the importance of including a thematic map with all the main faults. The latter can be mapped by surface studies, and their importance at depth can only be estimated from their relation with the geophysical data that provide information on the resistivity of the deep strata. As some faults may have strayed at depth, the integration with resistivity data would provide a way to differentiate the sealed faults from those in the active areas. The apparent resistivity map used in this study was constructed from data obtained with a Schlumberger array with a 2 km separation between electrodes ($AB/2 = 1$ km). This electrode separation

was selected because 2-D modeling shows that it contains the information about most relevant features of the deep structure of the reservoir (Campos-Enriquez and Garduño-Monroy, 1995).

5. Input data (thematic maps)

Five data sets were obtained from the Mexican Federal Commission of Electricity (CFE) for the development of a GIS that would predict the best areas to locate exploration wells. The data introduced in the model are surface geology, resistivity surveys, surface fracturing mapping, surface manifestations, and main faults.

5.1. Geology

Geologic contacts of the most recent rhyolitic domes were digitized. It was assumed that they would be related to the intrusive bodies that might serve as the heat source of the hydrothermal system (Fig. 2).

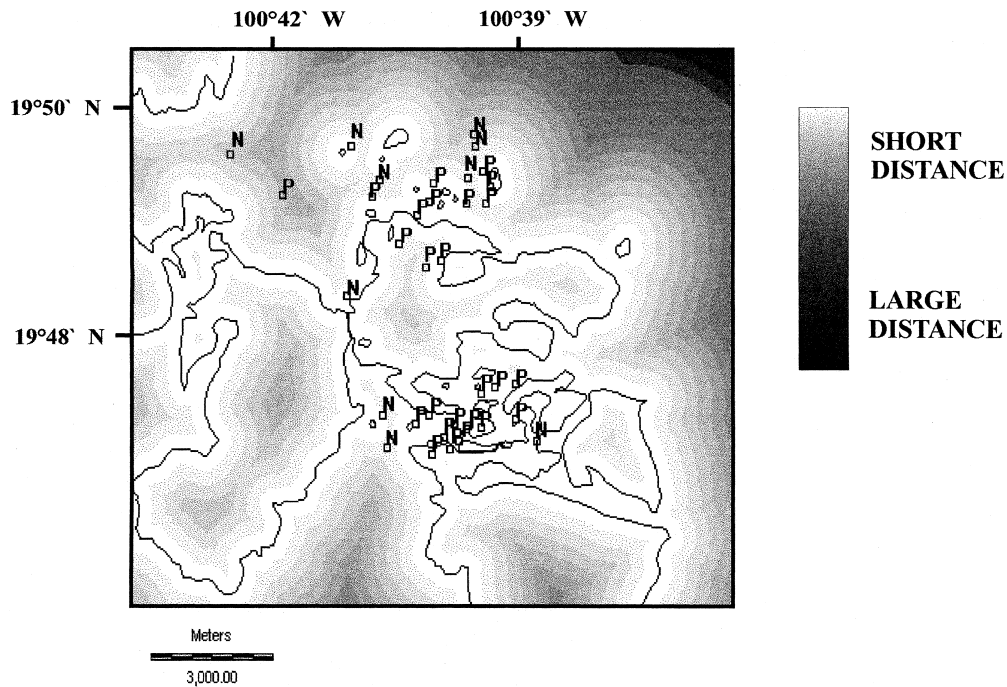


Fig. 2. Distance to contacts between surface manifestations and recent rhyolitic domes. P: producing wells; N: non-producing wells. Contacts are shown as black lines.

5.2. Structural map

Only the youngest faults were digitized: the E–W and N–S systems. Assuming that no well data are available, as would be the case in a reconnaissance study, all fault systems would be given the same weight. No difference between those systems was assumed because no surface evidence defines the prevalence of one of the fault systems on the permeability of the geothermal reservoir. It is assumed that a good model would derive the relative importance of each fault system on the basis of its relation with the other evidence (Fig. 3).

5.3. Surface manifestations

All the areas occupied by known surface manifestations, i.e., acid hot springs, fumaroles, steaming ground, and mud pools, were digitized and included in the map of surface manifestations. No different values were assigned to the different types of manifestations, as, according to the model of the field, all those features may be closely related to main upflows of the system at depth (Fig. 2).

5.4. Surface fracturing

This information was included under the assumption that the locally observed fracturing was related to strong stresses at depth related with vigorous

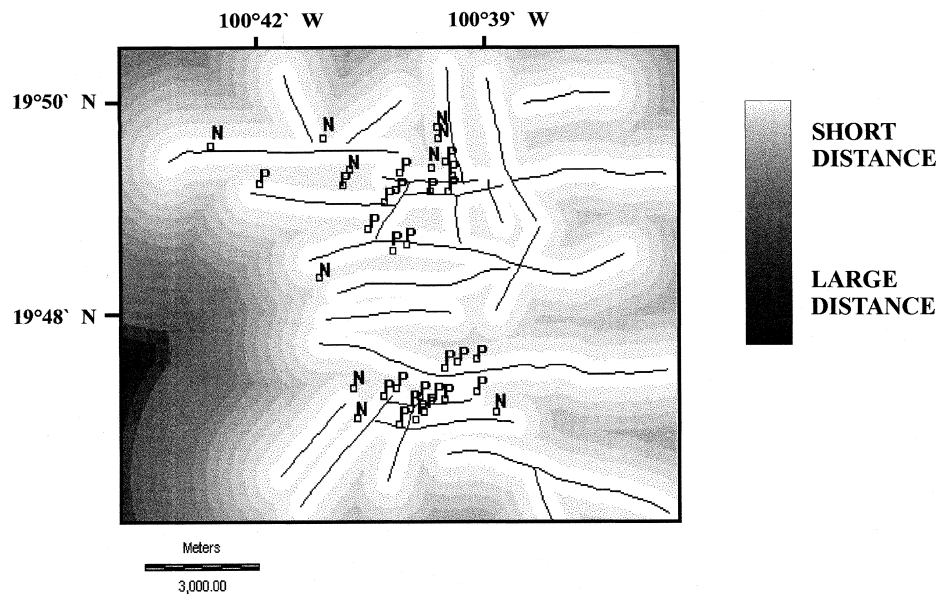


Fig. 3. Distance to main faults. P: producing wells; N: non-producing wells. Main faults are shown as black lines.

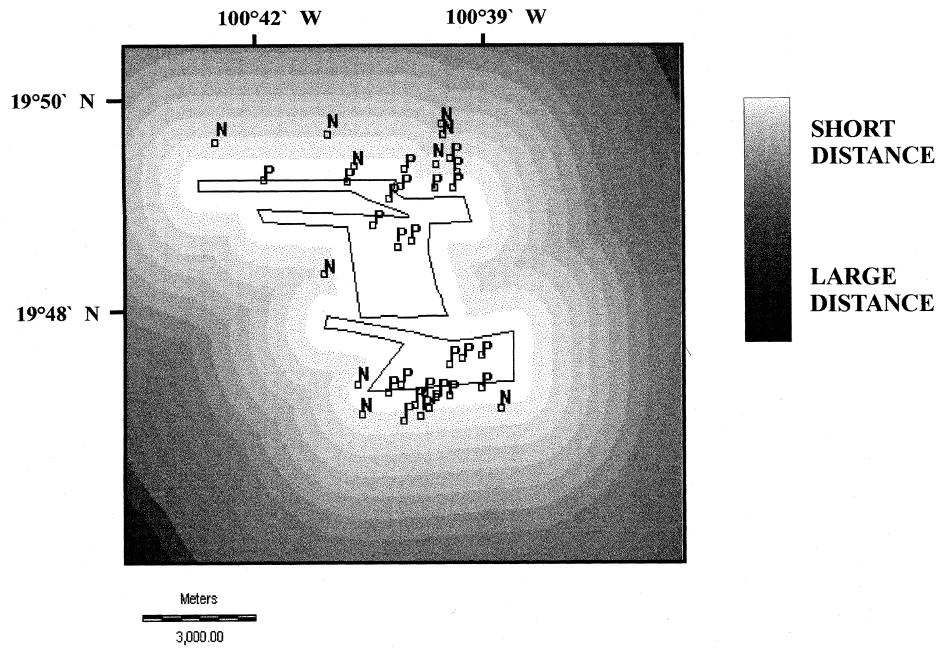


Fig. 4. Distance to areas with high surface fracturing mapped in Los Azufres geothermal field. P: producing wells; N: non-producing wells. The areas with a high density of surface fracturing (after López-Hernández, 1991) are shown as black line polygons.

hydrothermal activity (hydraulic fracturing?), and could indicate productive reservoir zones (Fig. 4).

5.5. Resistivity contours

The raster image was constructed on the basis of the map of apparent resistivity with contour values 10, 20, 50, 100, 200, 300, and 400 ohm m (Fig. 5).

Thematic maps in raster form were constructed from digitized maps of surface geology and resistivity data. The 30 m cell dimensions were chosen in order to facilitate further inclusion of remotely-sensed data. The maps described in Sections 5.1 to 5.5 were digitized as vector images including only lines, and were transformed to raster form using the raster/vector conversion in IDRISI. Rasterization of the resistivity map (Fig. 5) was performed with the INTERCON command in the IDRISI¹ program, which performs a linear interpolation on a contour map (Eastman, 1997). The discrimination of the prospective areas is given on the basis of the distance to the important features, so that the input “distance”

¹ The reference to commercial software does not imply the endorsement by the author and is meant only as a specification of the method utilized when processing the data.

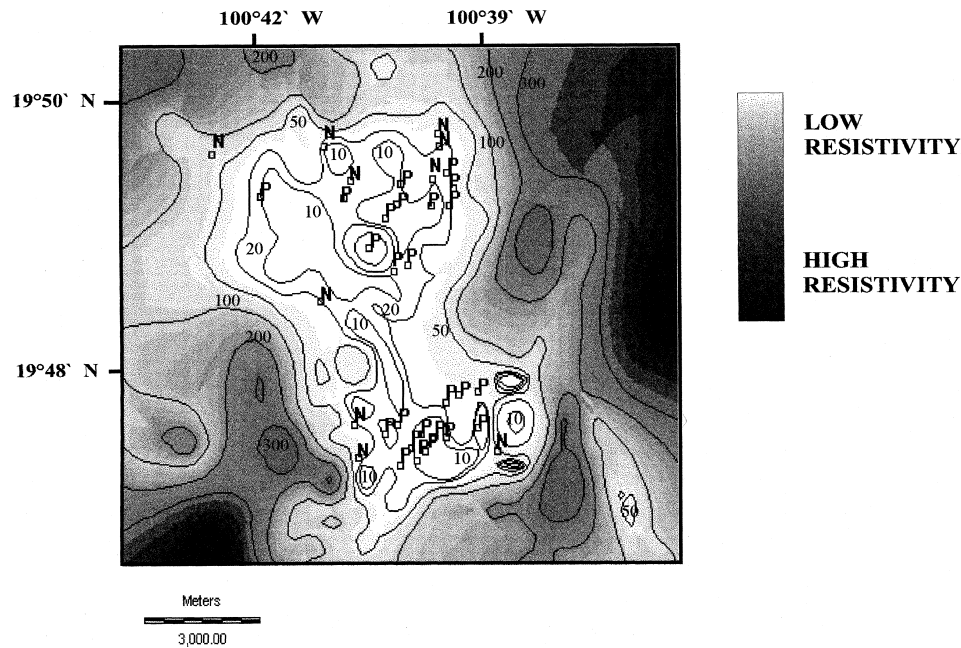


Fig. 5. Resistivity anomalies measured in Los Azufres geothermal field (Schlumberger array, AB/2 = 1 km) (after García, 1995). P: producing wells; N: non-producing wells. Contour values are given in ohm m.

images for rhyolitic domes contacts and surface manifestations (Fig. 2), main faults (Fig. 3), and surface fracturing (Fig. 4) were derived from the thematic maps in raster form using the **DISTANCE** command in the IDRISI program. This command calculates the minimum distance from the pixel to the closest feature in the image (Eastman, 1997).

6. GIS and data integration modeling

In applying data integration models to exploration of natural resources, expertise is important in the selection of the maps that will provide predictor keys of the resource to be studied, and if possible to avoid the cost of gathering redundant information. Also, the assignment of weights (in the Index Overlay model) and the relative importance of the evidence (in the Fuzzy data analysis) follow empirical relationships that depend on the conceptual model of the system to be studied. Firstly, the selection should be made between data-driven and knowledge-driven models. Data-driven models process the input data using weights of evidence, logistic regression or neural network analysis, and require training data to calculate the model parameters. Knowledge-driven models include model parameters that are estimated by an expert using fuzzy data analysis, or

Bayesian probability (Bonham-Carter, 1994; Bonham-Carter et al., 1994). The aim of this work is to provide evidence on the applicability of GIS to defining the most likely areas for producing geothermal energy on the basis of data collected during the reconnaissance stage of exploration. Location of favorable localities in an unexplored geothermal field assumes that no wells have been drilled, and that no data are available about the location of the production zones in that field; therefore, data-driven models are not used and only knowledge-driven models will be applied to this problem.

Three knowledge-driven models are proposed: Boolean logic model, Index Overlay model and the Fuzzy data analysis. In all models the same input data are used and the same conditions are to be satisfied, i.e. the production areas are assumed to be located:

1. in close relation to recent rhyolitic domes (at a distance of less than 100 m) OR
2. in the vicinity of the main upflows of the system marked by surface manifestations (at a distance of less than 100 m), OR
3. in the vicinity of active faults (at a distance of less than 100 m), OR
4. within or at a distance of less than 100 m from areas that present a high density of surface fracturing according to structural studies (López-Hernández, 1991) AND
5. in areas where resistivity is less than 10 ohm m.

These conditions were selected, assuming that the field corresponds to the conceptual model defined from the geological data and the geochemical characteristics of the surface manifestations. The characteristics of a geothermal reservoir are typically high permeability and higher-than-normal geothermal gradient. In the case of Los Azufres geothermal field, permeability is related to the degree of fracturing of the reservoir rocks. The location of main faults and the high density of surface fracturing are very useful surface observations in the determination of secondary permeability. The estimated 100 m as favorable distance is based on the observations of high permeability areas in fractured systems used in hydrological studies (Gustaffson, 1993). The 100 m distance is smaller than the estimation of the favorable distance to faults used in mineral deposits, where distances of almost one kilometer are considered favorable, to take into account the possibility of an inclined plane (Bonham-Carter, 1994). This is not the case of Los Azufres, where the main faults have large dip angles (Huitron et al., 1991). Transport of the fluid from deeper parts of the reservoir to the surface is evidenced by surface manifestations such as hot springs, fumaroles and hot pools. Surface observations in Los Azufres can be used to delineate areas characterized by high permeability and hot fluid transport from the reservoir. These areas can be outlined in the resulting image after the proper combination of the thematic maps that contain the main faults, the fractured areas, the surface manifestations and the boundaries of the most recent rhyolitic domes. Conditions at depth related with producing zones are characterized by low apparent resistivity, typically less than 10 ohm m. The simultaneous occurrence of low resistivity at depth and surface features indicating high permeability and fluid

upflow may be evidence of the presence of a reservoir. These areas were obtained by the OR operation applied to the thematic maps that contain the distance to the recent rhyolitic domes, main faults, surface fracturing and surface manifestations, combined (AND operation) with the map that included the areas within the 10 ohm m contour, according to the inference network given in Fig. 6. These operations can be represented also by the formula:

$$\text{Resistivity} \cap ((\text{faults} \cup \text{fracturing}) \cup (\text{rhyolite} \cup \text{manifestations})) \quad (1)$$

The results obtained with these operations are expressed in a scale from 0 to 1, where 0 indicates the less favorable areas and 1 the most favorable areas. This scale does not strictly represent probability; it is therefore more conveniently denominated as a favorability scale.

6.1. Boolean logic model

Boolean images were constructed on the basis of the input thematic maps taking into account the conditions assumed for the GIS model (1–5). For each map, the areas where the conditions were fulfilled were assigned a value of 1 while the rest of the pixels had a value of 0. Then the logic operations (AND, OR) were performed using the OVERLAY command of the IDRISI program according to the scheme for data integration shown in Fig. 6. The final result (Fig. 7) yields an image in which the areas where all conditions are fulfilled are presented in black. Fig. 7 also shows the location of wells that have been considered as “production

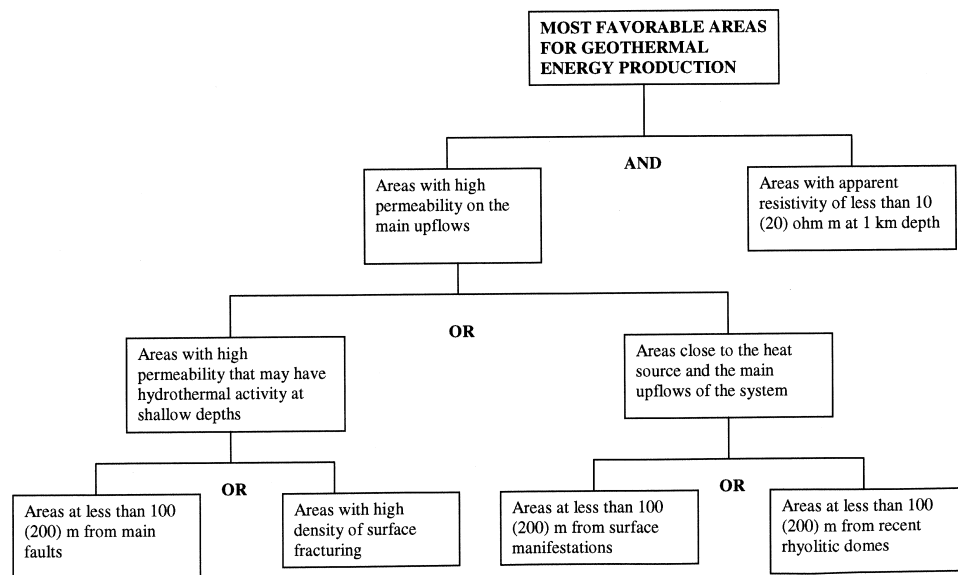


Fig. 6. Proposed inference network for data integration in geothermal exploration.

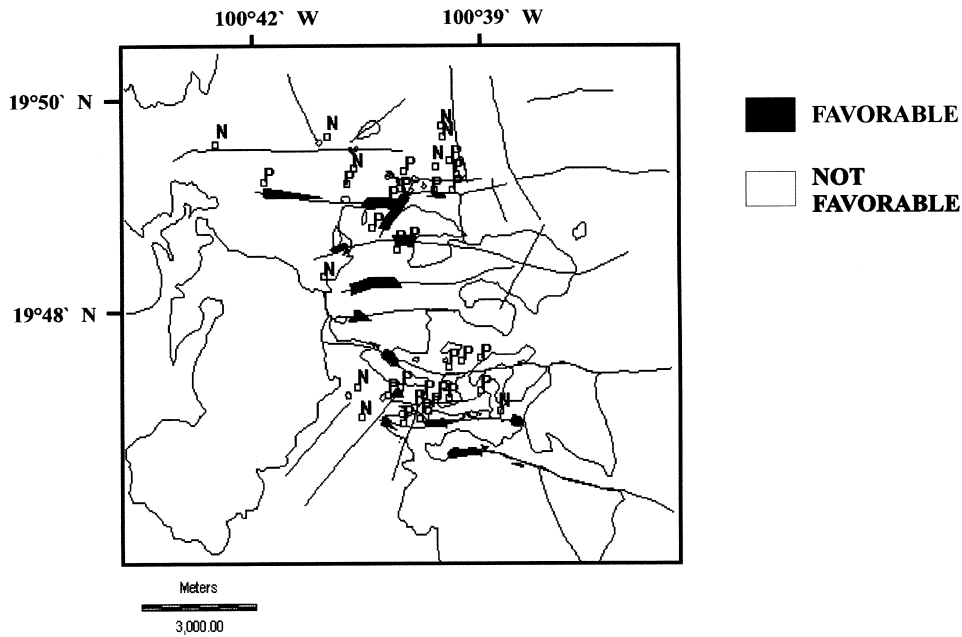


Fig. 7. Favorable areas for geothermal energy production as identified by the Boolean model. P: producing wells; N: non-producing wells. Surface manifestations, contacts with young rhyolitic domes, and faults are also shown.

wells”, the main faults and the contacts with young rhyolitic domes and surface manifestations. Favorable areas in this model include only production wells and all non-producing wells were left out; however, many production wells are located outside the marked favorable areas, indicating that the Boolean model has omission errors that might be overcome using a more flexible combination of the evidence maps. A more flexible approach is proposed in the following method: the index overlay with maps that include three classes and intermediate values can be considered relatively favorable areas.

Table 1

Weights assigned to the thematic maps for geothermal exploration, and scores of the three classes used in the Index Overlay model

Weight	Thematic map	Conditions	Score
5	Rhyolitic domes	Distance ≤ 100 m, > 100 & ≤ 200 m, > 200 m	10, 8, 0
5	Surface manifestations	Distance ≤ 100 m, > 100 & ≤ 200 m, > 200 m	10, 8, 0
5	Surface fracturing	Distance ≤ 100 m, > 100 & ≤ 200 m, > 200 m	10, 8, 0
7	Main faults	Distance ≤ 100 m, > 100 & ≤ 200 m, > 200 m	10, 8, 0
9	Apparent resistivity	≤ 10 ohm m, > 10 & ≤ 20 ohm m, > 20 ohm m	10, 8, 0

6.2. Index Overlay model

The Index Overlay model uses the same thematic maps as the Boolean model, but multiplying them by weight factors with different scores that yield three classes instead of two (Table 1). The weight factors are assigned to each evidence map according to its relevance for determining the presence of the geothermal reservoir (Table 1). An expert evaluates the relevance of the data according to the relation between the evidence and permeability, geothermal fluid presence and the heat source location, and defines the scores assigned to different classes. In this case the scores and weights are chosen as integers in a scale from 0 to 10 for all maps. The resulting image is obtained using Eq. (2), which calculates the average score for each pixel:

$$S = \frac{\sum_i^n S_{ij} w_i}{\sum_i^n w_i} \quad (2)$$

where S is the weighted score for each pixel, w_i is the weight for the i th thematic

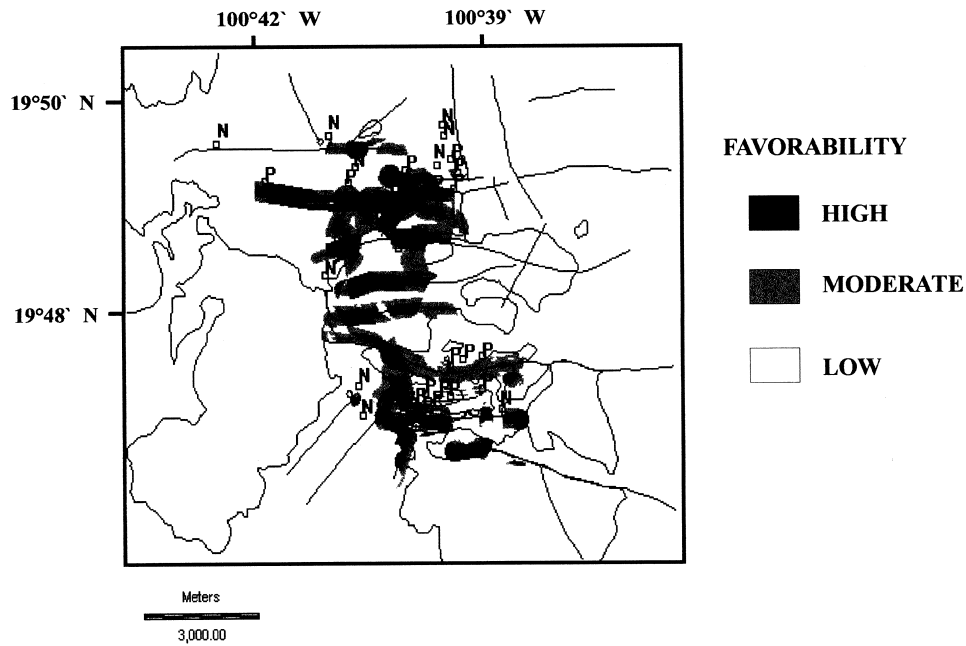


Fig. 8. Result of the Index Overlay modeling using the weights given in Table 1 for each evidence binary map. P: producing wells; N: non-producing wells. High favorability areas indicate resulting scores of more than 0.7, moderate favorability ranges from 0.6 to 0.7, and low favorability areas have scores of less than 0.6. Surface manifestations, contacts with young rhyolitic domes, and faults are also shown.

map, and S_{ij} is the score for the j th class of the i th thematic map (Bonham-Carter, 1994).

As shown in Table 1, the same thematic maps included in the Boolean analysis are used in this model. The same weight factor (5) was assigned to the thematic maps that show the presence of young rhyolitic domes, surface manifestations and high fracture density on the surface, while the main faults and apparent resistivity maps were assigned higher weights, 7 and 9, respectively. Three classes were taken into account in each map: the areas with a distance to main features of 100 m would have a score of 10, the areas within a distance between 100 and 200 m were assigned a somewhat lower score (8), which is still high enough to allow their inclusion as favorable areas for production. The areas at a distance of more than 200 m were assigned a score of 0. The 200 m distance was chosen on the basis of the recommendation of the field engineers, who observed that the production wells are usually located within that distance from the main faults. The favorable low-resistivity area was enlarged to include areas with apparent resistivity between 10 and 20 ohm m. A score value of 10 was allocated to areas with apparent resistivity less or equal to 10 ohm m, and a score value of 8 for values between 10 and 20 ohm m; the areas with apparent resistivity higher than 20 ohm m were assigned a score of 0.

The resulting image, obtained with the Index Overlay Technique with Multi-class Maps (Bonham-Carter, 1994), has values from 0 to 1. Fig. 8 shows the favorable areas with values higher than 0.6, which is the lowest value that includes most production wells; in order to enhance the visualization of the favorable areas, values lower than 0.6 were not displayed. Areas with values higher than 0.7 leave most production well zones out, as in the case of the results obtained with the Boolean model. It is important to note that in the Index Overlay model, again the non-producing wells are not included in the marked favorable areas.

6.3. Fuzzy data analysis (Multi-Attribute Decision Making)

Fuzzy classes are sets without sharp boundaries, that is, the transition between membership and non-membership of a location in the set is gradual (Zadeh, 1965). A fuzzy set is characterized by a fuzzy membership grade that ranges from 0.0 to 1.0, indicating a continuous increase from non-membership (0.0) to complete membership (1.0). For instance, in the case of distance to main faults,

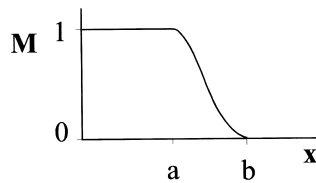


Fig. 9. Fuzzy membership function (M) used in the fuzzy data analysis; a is the point where membership function falls below 1, and b is the point where membership function becomes 0.

complete membership (membership value = 1) is assigned to pixels at a distance of 100 m or less, then the fuzzy membership grade decreases according to a decreasing cosine function up to 200 m (Eq. 3). Pixels located at distances greater than 200 m are assigned non-membership grade (membership value = 0). The shape of the function is shown in Fig. 9.

$$M = \cos \alpha \quad (3)$$

In the case of a monotonically decreasing function: when $x < \text{point a}$, $M = 1$; when $x > \text{point b}$, $M = 0$; when $\text{point b} \geq x \geq \text{point a}$, M is calculated using Eq. (3) and α is defined as $(x - \text{point a}) / (\text{point b} - \text{point a}) * \pi / 2$ (Fig. 9). In the case of the thematic maps of the distances to features, a is 100 and b is 200 m; and $a = 10$ and $b = 20$ ohm m in the case of the resistivity map.

Fuzzy images were obtained for the five thematic maps of Los Azufres geothermal field with a monotonically decreasing function such as that shown in Fig. 9. The inflexion points were input as described above, and the fuzzy membership images were obtained using the FUZZY command in the IDRISI program.

The decision-making process involves a choice between alternatives. In the case of geothermal exploration, the alternatives represent different hypotheses about the deep structure of a geothermal reservoir and the best locations on the surface to site wells so as to reach areas with favorable conditions for geothermal energy production. Therefore, all the pixels in the study area will be characterized according to their fuzzy membership grade assigned on the different thematic maps used as input in the model.

The decisions are taken on the basis of criteria, which in our case are parameters such as geologic contacts, presence of active faults, low resistivity, following a procedure that is called the decision rule. In decision-making, Multi-Criteria Evaluation must determine the optimal alternative with the highest degree of desirability with respect to all relevant goals (Zimmermann, 1985, 1996), which can be achieved by hierarchical aggregation procedures (Saaty, 1977). This approach is based on the pair-wise comparison of alternatives with respect to each

Table 2

Matrix of the relative importance between pairs of thematic maps used to calculate the individual weights in the Fuzzy data analysis

	Resistivity	Main faults	Surface fracturing	Rhyolitic domes or surface manifestations	Calculated weights
Resistivity	1				0.3899
Main faults	1	1			0.3899
Surface fracturing	1/5	1/5	1		0.0679
Rhyolitic domes or surface manifestations	1/3	1/3	3	1	0.1523

criterion. The comparison concerns the relative importance of each two criteria in determining suitability for the stated objective. In IDRISI, the ratings are on a 9-point scale: 1/9, 1/7, 1/5, 1/3, 1, 3, 5, 7, and 9 that range from extremely less important to extremely more important. The matrix of the pair-wise comparison of the criteria in the thematic maps is shown in Table 2. The principal eigenvector of this matrix is used to calculate the best fit of weights. This is done with the **WEIGHT** command in IDRISI. The weights obtained are given in the last column of Table 2.

The weights and the calculated fuzzy images are input in the Multi-Criteria Evaluation (MCE) operator, which combines the evidence maps in order to calculate the suitability or favorability of each area element. This is performed through an ordered weights combination, which multiplies the score of the evidence in each element by the weight and the order weight. The order weight is assigned to the evidence in relation with the degree of risk that may be taken in the decision process; in the case of a low-risk decision, a high order weight must be assigned to the evidence with higher weight in the pair-wise comparison process. The result is the same as that of applying the Fuzzy AND operator, which minimizes the favorability value assigned to each area element; the operator considers each evidence as necessary but not sufficient to yield a high favorability value. When a high-risk decision is acceptable, a high order weight is assigned to evidence with a low pair-wise comparison weight; this calculates the Fuzzy OR operator that maximizes the result. This operator considers any evidence as sufficient and will therefore facilitate the inclusion of an area element as favorable in the resulting favorability map.

The order weights may be varied with the restriction that their sum must not be greater than one. Where the order weights have the same value for each evidence map, the result is a weighted linear combination of the evidence.

Here we present only the results obtained with the high- and low-risk combination of ordered weights, as they show the results obtained in both extremes of the Multi-Criteria Evaluation. The results are presented in Figs. 10 and 11.

The low-risk image is similar to the result of the Boolean analysis; many producing wells are left out of the areas marked as favorable, and none of the non-producing wells are included. This shows that the criteria and relative weights assigned in the fuzzy analysis are concordant with the conditions (1–5) established in the GIS model. The high-risk image includes as favorable all pixels that satisfy any restrictive criteria; it can, therefore be noted that although all producing wells are contained in the marked areas, the non-producing wells are also included.

7. Discussion

In mineral exploration as well as in geothermal exploration, the emphasis is laid on structural and geologic features, and on geochemical data (Campbell et al., 1982; Agterberg, 1989; Bonham-Carter et al., 1988; Chung et al., 1992;

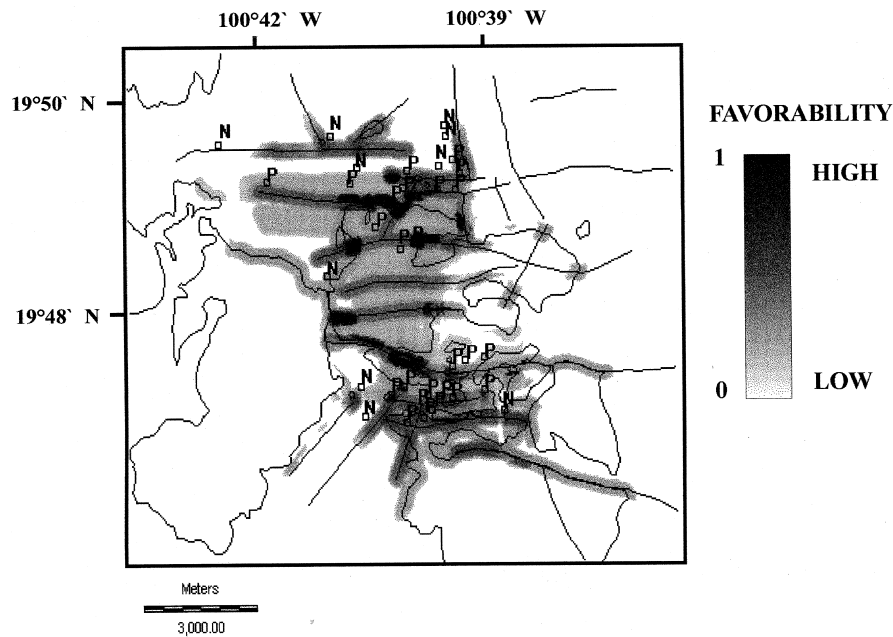


Fig. 10. Image of high-risk areas for geothermal energy production, obtained using the Fuzzy Logic model. P: producing wells; N: non-producing wells. Surface manifestations, contacts with young rhyolitic domes, and faults are also shown.

McCammon, 1993; Bonham-Carter et al., 1994). In both cases, the key evidence used is similar, so that the digital integration framework of geoscientific data developed in the mineral industry can be adapted to geothermal resources exploration.

In geothermal exploration, the areas to be identified are those in which geothermal fluids can be found at temperatures higher than 200°C, and permeability is high enough to allow the inflow of geothermal fluid from the reservoir to the vicinity of the well. The requirements for those areas are given in the inference network (Fig. 6) presented here for a geothermal field located in rough terrain and with permeability parameters related to fracturing of the reservoir rock.

The similarity of the results obtained with the Boolean and the Low-Risk models shows that the inference network used as the basis of the model and the relative importance of the evidence used in the assignment of the weights in the Multi-Criteria Evaluation are in good agreement.

After comparison with the well production data (Table 3), the Boolean model proved to be too restrictive, as the areas marked as favorable are those in which all criteria in the inference network are satisfied. As a result of these restrictions, many production wells were left out of the favorable areas. It must, therefore be inferred that production wells can be located in areas in which not all the imposed

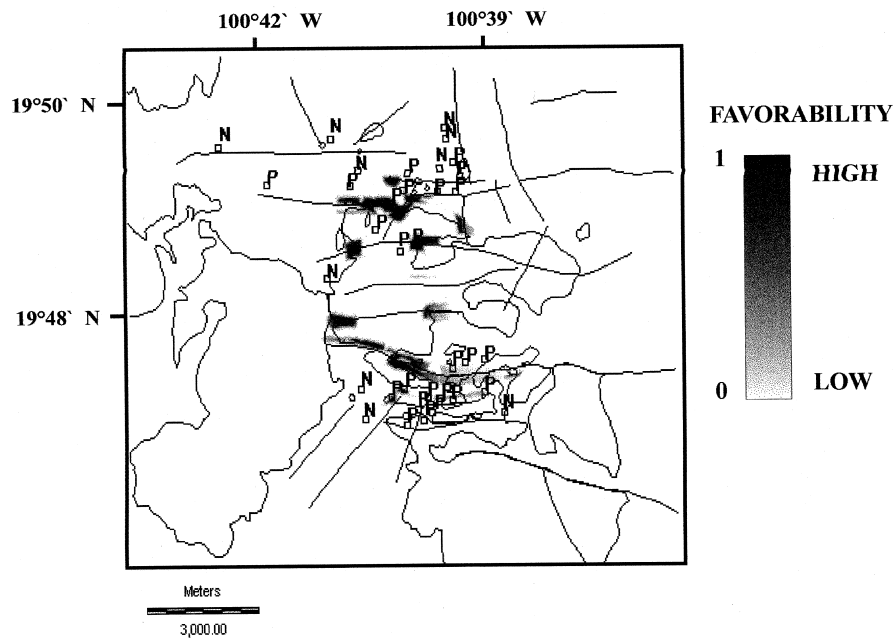


Fig. 11. Image of low-risk areas for geothermal energy production, obtained with the Fuzzy Logic model. P: producing wells; N: non-producing wells. Surface manifestations, contacts with young rhyolitic domes, and faults are also shown.

Table 3

Number of production and non-production wells located in areas marked as most, moderate and least favorable in the three models applied

	Most favorable		Moderate		Least favorable	
	#P	#N	#P	#N	#P	#N
Boolean ^a	7	0	—	—	18	10
Index Overlay ^b	12	0	10	0	3	10
Fuzzy High Risk ^c	4	1	20	4	1	5
Fuzzy Low Risk ^d	8	0	6	0	11	10

^a Only two classes included: 1 and 0.

^b Three classes included; weights for each thematic map are given in Table 1. Scores for distances ≤ 100 m; $> 100 \& \leq 200$ m; > 200 m are 10, 8, and 0 respectively.

^c Fuzzy membership function shown in Fig. 9. Integration scheme involves fuzzy union operations (maximizing operator).

^d Fuzzy membership function shown in Fig. 9. Integration scheme involves fuzzy intersection operations (minimizing operator).

criteria are fulfilled, and that more flexible models have to be developed that will identify areas that do not satisfy all requirements but are still favorable for geothermal energy production. However, an important feature of this model is that no failed wells fall within the favorable areas (Table 3).

It is important to note that when a decision has to be taken about drilling an exploration well, only the most favorable areas must be considered. The Boolean or the Low-Risk models are therefore more suitable for exploration well siting.

The Index Overlay and the High-Risk models lessen the requirements for each pixel and the decision rule, with the result that they do not have to satisfy all the requirements set by the Boolean or the Low Risk models to be considered as relatively favorable areas. However, as shown in the comparison with the production wells location, values higher than 0.6 in the favorability scale from 0 to 1 in the Index Overlay model denote an area that includes all drilled production wells in Los Azufres geothermal field, and leave out all non-producing wells. This model can provide helpful data for planning further drilling, and can also be used to plan detailed exploration. The High-Risk model delineates a large area that includes non-producing wells and can therefore only be used to plan further detailed pre-drilling.

Frequently, production wells are located within the low resistivity anomalies, at the intersection of main faults and at a short distance from the surface manifestations. However, the results show that more areas may be suitable for geothermal exploitation, and a GIS model should be constructed to take into account all the evidence, in order to define all areas likely to be exploited without overlooking areas with an important geothermal potential. The integration of all the available data in a GIS provides an important tool for the evaluation of geothermal fields in different exploration and development stages.

8. Conclusions

In the early exploration stages of areas where secondary permeability predominates, preliminary siting of exploration wells can be achieved by means of a geographical information system based on surface geology and resistivity surveys. This system can also be used to identify areas suited to more detailed exploration before drilling an expensive exploration well; for instance, costly detailed geothermal gradient measurements could be performed only in the areas that the Index Overlay model has assigned favorability values higher than 0.6.

In Los Azufres a good agreement was obtained between the location of the production wells and the areas delineated by Boolean, Index Overlay and Fuzzy data analysis models on the basis of available geoscientific data.

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